

Applied Data Science Capstone

SpaceX Falcon 9 First Stage Landing Prediction

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IBM Data Science Professional Certificate

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Executive Summary

Methods Used

- Data collection via SpaceX REST API and Wikipedia web scraping
- Data wrangling and labeling of landing outcomes
- Exploratory Data Analysis with visualization and SQL
- Interactive visual analytics with Folium and Plotly Dash
- Predictive analysis using four classification models, tuned with GridSearchCV

Key Results

- Launch success rate climbed from 0% (2010-2013) to ~85-90% by 2019-2020
- KSC LC-39A is the top-performing site: 10 successes, 76.9% success rate
- Payloads in the 2,000-5,500 kg range land most reliably
- Orbits ES-L1, GEO, HEO, and SSO show a 100% success rate
- Decision Tree classifier was the best model, with 87.5% cross-validated accuracy

Introduction

Project Background

SpaceX advertises Falcon 9 launches at roughly \$62 million, far below the \$165 million+ charged by competitors. Much of that savings comes from reusing the rocket's first stage. If the first stage's landing outcome can be predicted from public launch data, the cost of a launch can be estimated in advance — useful information for any company considering bidding against SpaceX.

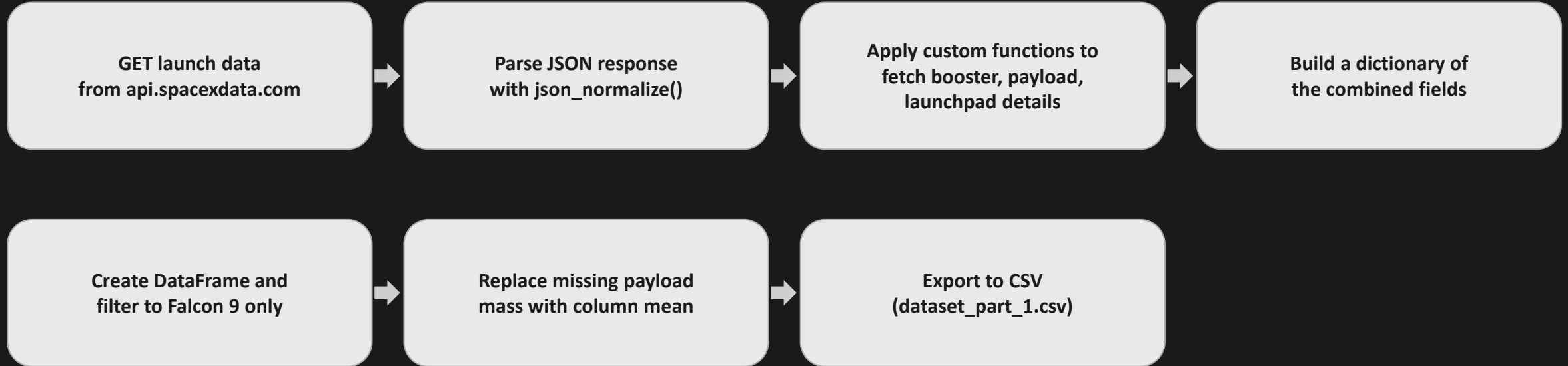
Questions to Answer

- How do payload mass, launch site, flight number, and orbit type affect first-stage landing success?
- Has the landing success rate improved over time?
- Which classification algorithm best predicts landing outcome for this dataset?

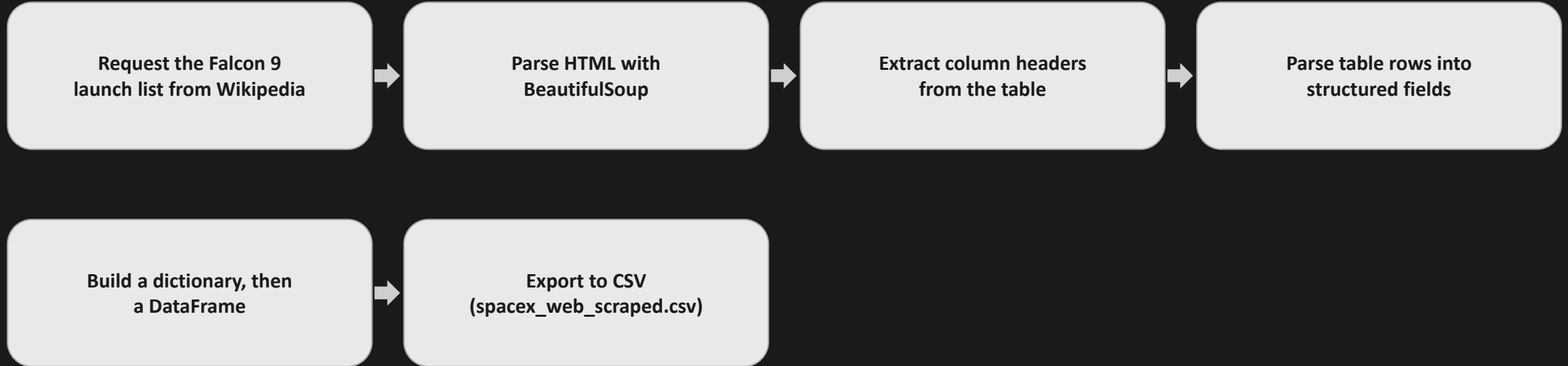
Methodology

- Data collection: SpaceX REST API requests and Wikipedia web scraping (BeautifulSoup)
 - Filtering to Falcon 9 launches and imputing missing payload mass with the column mean
- Data wrangling: derived a binary landing Class label from the raw Outcome column
- Exploratory Data Analysis using Matplotlib/Seaborn visualizations and SQL queries
- Interactive visual analytics with a Folium map and a Plotly Dash web app
- Predictive analysis: standardized features, train/test split, and GridSearchCV (cv=10) tuning of
 - Logistic Regression, SVM, Decision Tree, and KNN classifiers

Data Collection: SpaceX REST API



Data Collection: Web Scraping



Data Wrangling Methodology

Landing outcomes in the raw data include several success/failure variants — True/False Ocean, RTLS (return to launch site), and ASDS (drone ship). These were consolidated into a single binary training label: Class = 1 for any successful landing, Class = 0 for any failure or no attempt.

- Calculated launch counts per site
- Calculated occurrence of each orbit type
- Cross-tabulated mission outcome by orbit type
- Derived the binary landing Class label from the Outcome column
- Exported the cleaned dataset to CSV (dataset_part_2.csv)

EDA with Data Visualization

Charts produced:

- Flight Number vs. Payload Mass, Flight Number vs. Launch Site, Payload Mass vs. Launch Site
- Success Rate vs. Orbit Type, Flight Number vs. Orbit Type, Payload Mass vs. Orbit Type
- Success Rate Yearly Trend

Scatter plots reveal relationships between numeric variables that may be useful features for the classification model. Bar charts compare success rate across discrete categories (launch site, orbit). The line chart shows how the success rate trends over time.

EDA with SQL

- Unique launch site names
- Records where the launch site begins with 'CCA'
- Total payload mass carried for NASA (CRS) missions
- Average payload mass for booster version F9 v1.1
- Date of the first successful ground-pad landing
- Boosters with a successful drone-ship landing carrying 4,000-6,000 kg payloads
- Total count of successful vs. failed mission outcomes
- Booster version(s) that carried the maximum payload mass
- 2015 drone-ship landing failures, by booster and launch site
- Ranked count of landing outcomes between 2010-06-04 and 2017-03-20

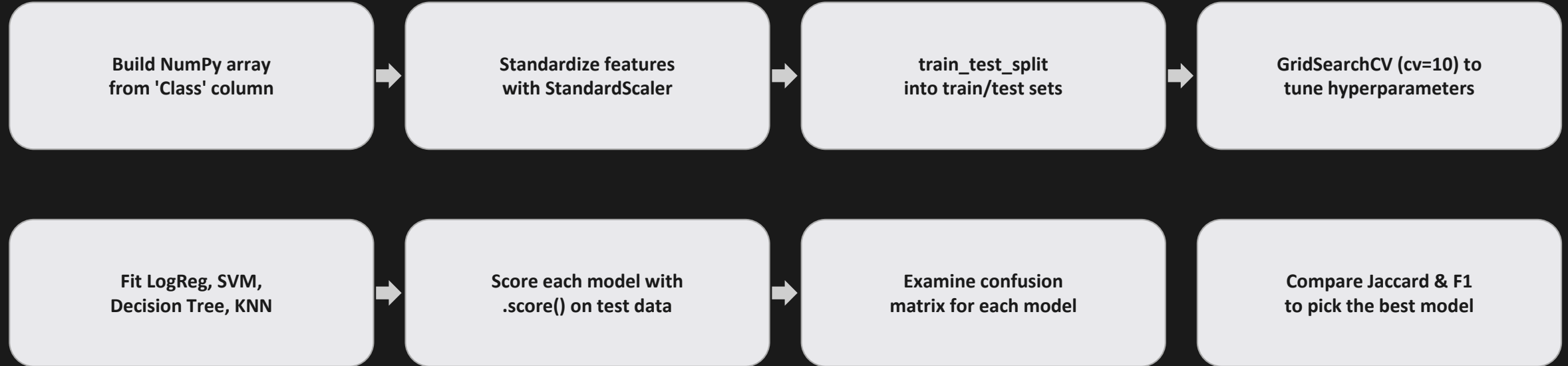
Interactive Visual Analytics: Folium Map

- Added a marker for NASA Johnson Space Center as a reference start location
- Added a marker (with circle + label) for each of the 4 launch sites, using lat/long
- Color-coded individual launch outcomes per site using MarkerCluster — green = success, red = failure
- Drew distance lines from KSC LC-39A to the nearest railway, highway, coastline, and city

Interactive Visual Analytics: Plotly Dash App

- Dropdown list to select a launch site (or All Sites)
- Pie chart of successful launches — all sites, or success/failure split for one selected site
- Range slider to filter by payload mass
- Scatter chart of payload mass vs. launch outcome, colored by booster version

Predictive Analysis: Methodology

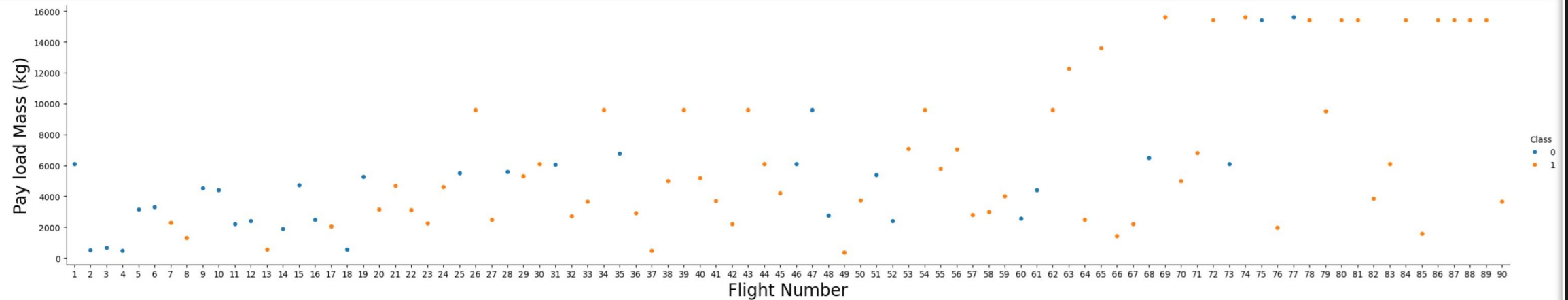


Results

EDA · Interactive Analytics · Predictive Analysis

Flight Number vs. Payload Mass

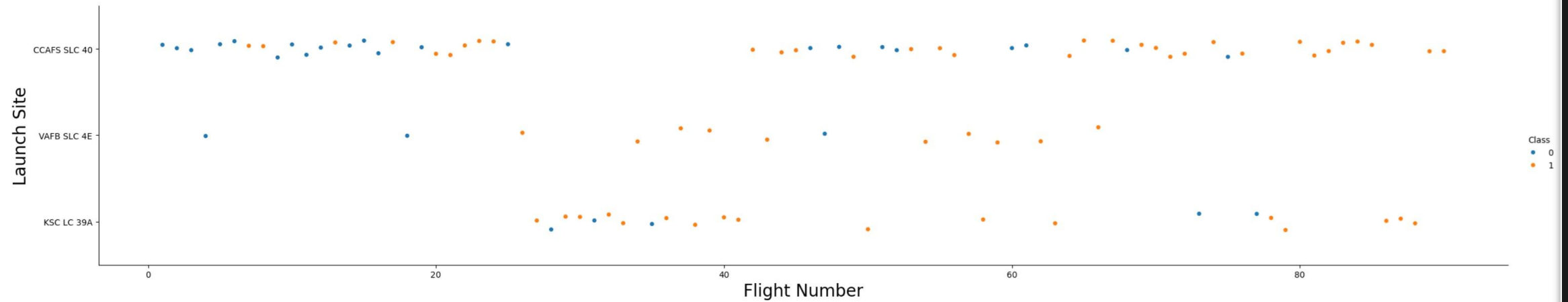
- Newer flights (higher flight number) tend to carry heavier payloads
- Later flights skew toward successful landings (orange) regardless of payload size
- Suggests SpaceX's landing technique matured over time, independent of payload mass



Flight Number vs. Launch Site

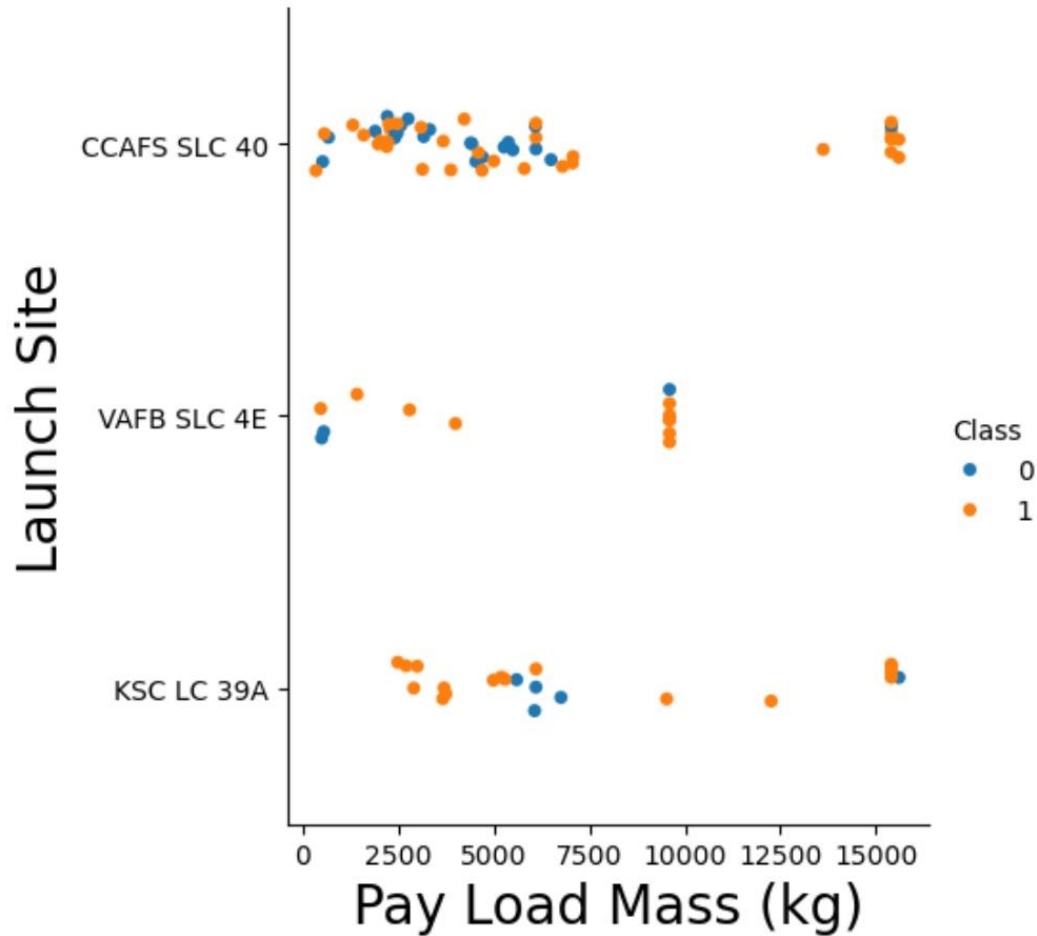
- The earliest flights mostly failed; the most recent flights mostly succeeded
- CCAFS SLC-40 hosts roughly half of all launches
- VAFB SLC-4E and KSC LC-39A show higher success rates
- Each successive launch is more likely to succeed than the last

```
[10]: Text(32.35652916666665, 0.5, 'Launch Site')
```



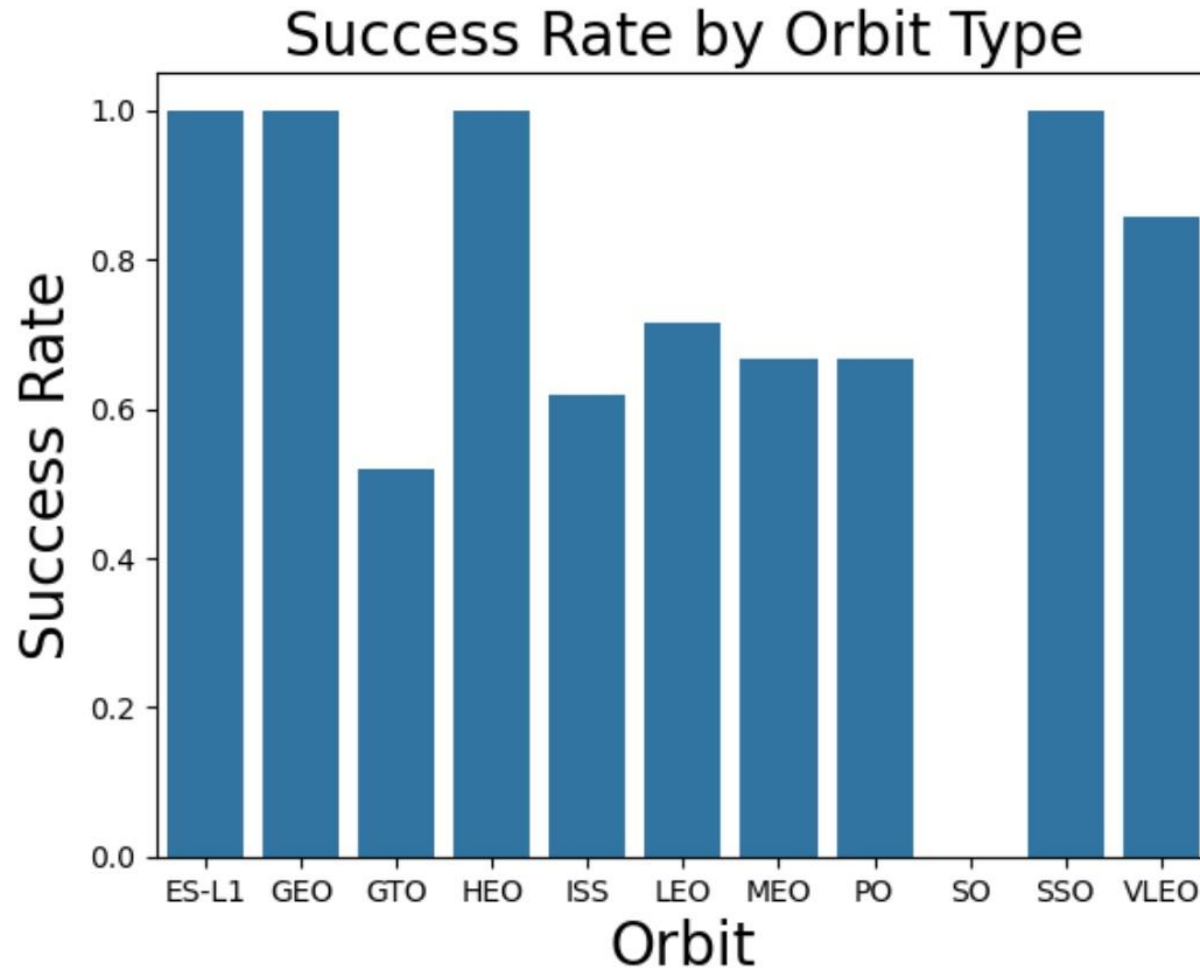
Payload Mass vs. Launch Site

```
[11]: Text(45.78264583333335, 0.5, 'Launch Site')
```



- At every site, heavier payloads correlate with a higher success rate
- Most launches carrying over 7,000 kg succeeded
- KSC LC-39A also achieves a 100% success rate for payloads under 5,500 kg

Success Rate vs. Orbit Type

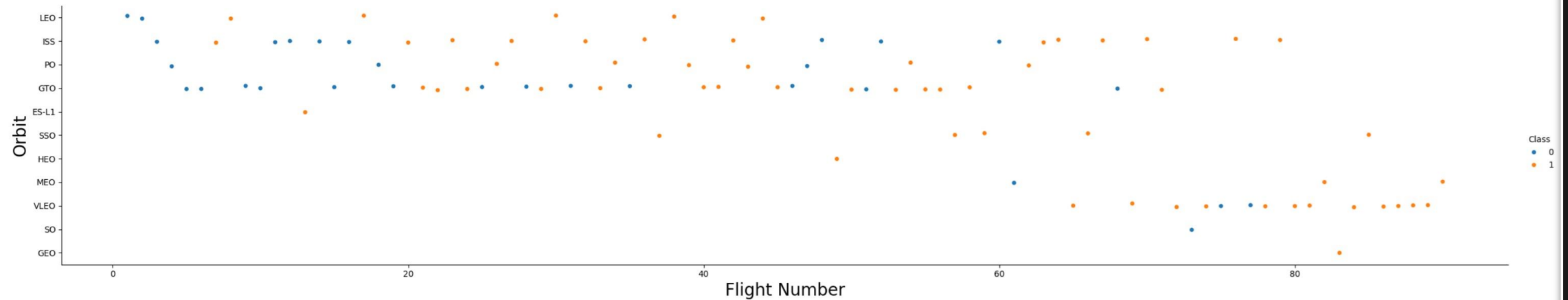


- 100% success: ES-L1, GEO, HEO, SSO
- 0% success: SO
- 50-85% success: GTO, ISS, LEO, MEO, PO
- Orbit type is a strong candidate predictive feature

Flight Number vs. Orbit Type

- In LEO, success appears related to flight number — later LEO flights succeed more often
- In GTO, flight number shows little to no relationship with success
- Indicates the learning curve effect is orbit-dependent

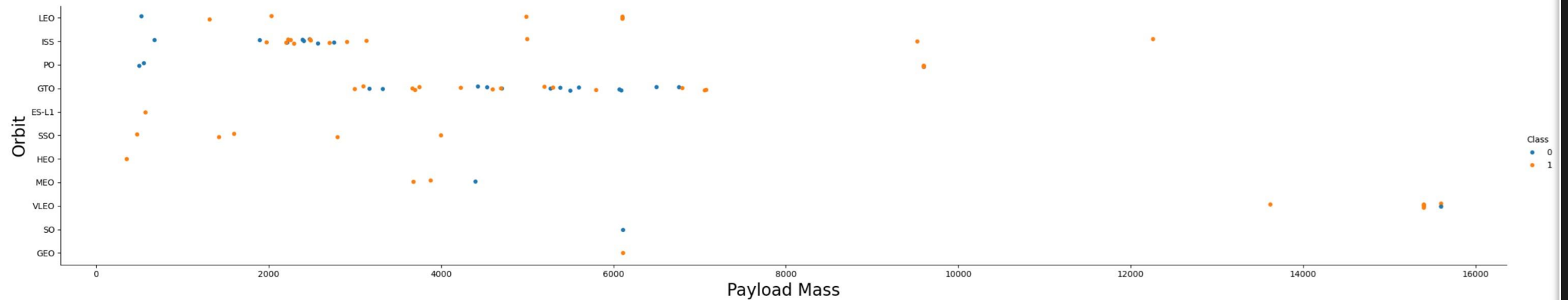
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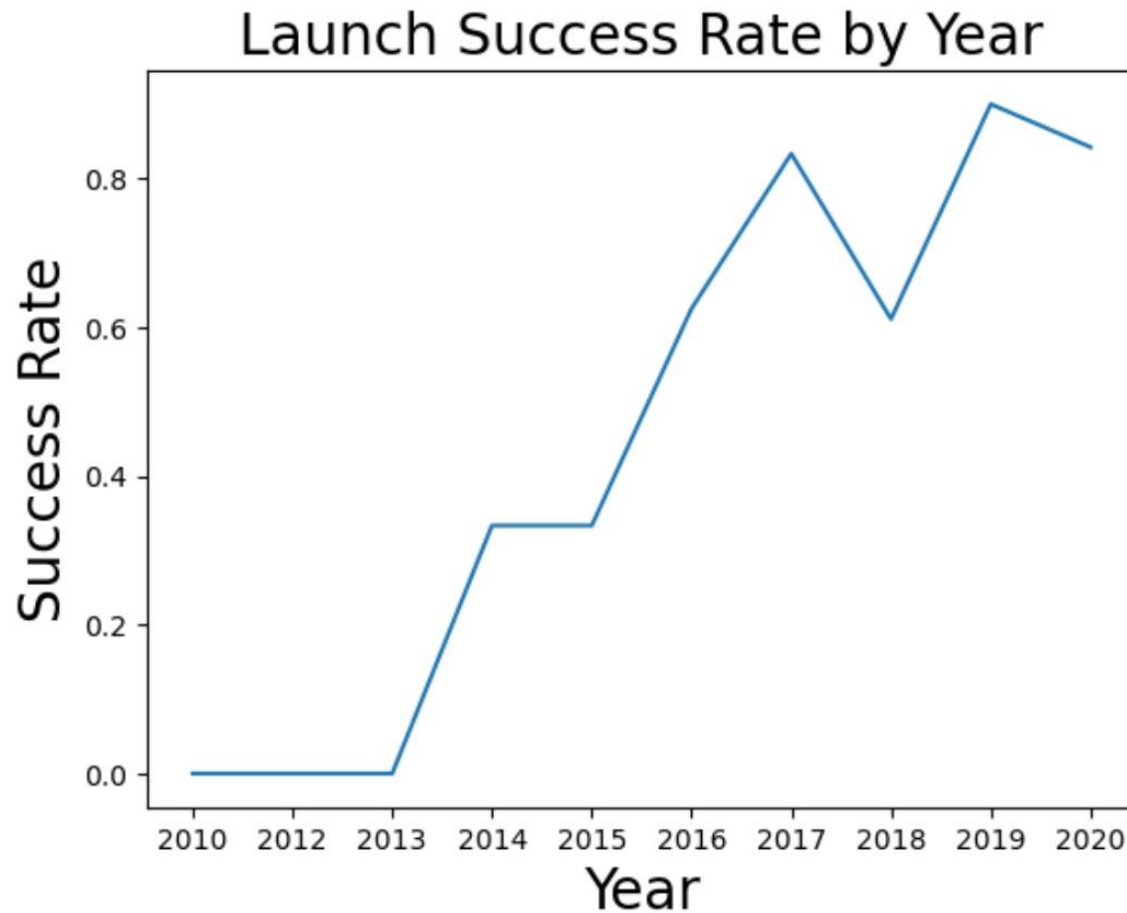
Payload Mass vs. Orbit Type

- Heavy payloads appear to hurt success in GTO orbit
- Heavy payloads appear to help success in LEO/ISS (Polar LEO) orbit
- Orbit and payload mass likely interact rather than acting independently

[14]: Text(30.961191666666664, 0.5, 'Orbit')



Launch Success Yearly Trend



- Success rate was 0% from 2010-2013
- Steady increase from 2013 through 2020
- Reflects SpaceX's iterative improvement of landing technique over time

EDA with SQL — Key Findings

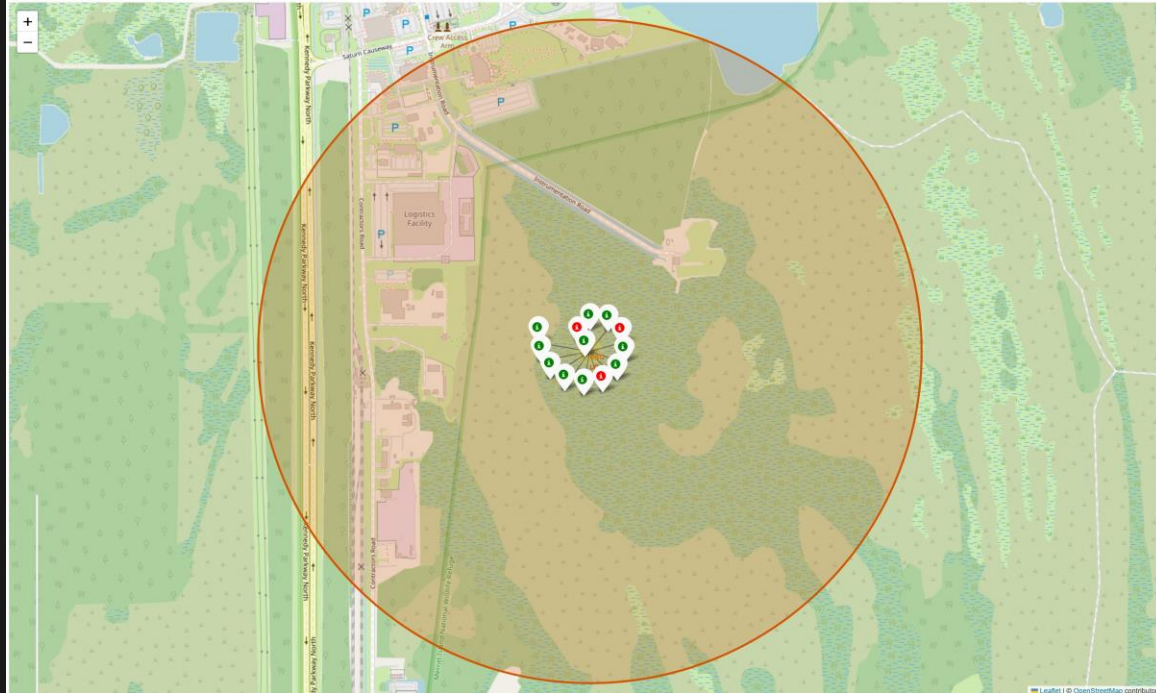
Query	Result
Unique launch sites	CCAFS LC-40, CCAFS SLC-40, KSC LC-39A, VAFB SLC-4E
Total payload mass for NASA (CRS)	48,213 kg
Average payload mass, F9 v1.1	2,534.7 kg
First successful ground-pad landing	2015-12-22
Successful drone-ship landings, 4,000-6,000 kg	F9 FT B1022, B1026, B1021.2, B1031.2
Total mission outcomes	Success: 98 Success (payload unclear): 1 Failure (in flight): 1
Booster carrying max payload mass	F9 B5 (multiple serials, e.g. B1048-B1060 series)
2015 drone-ship landing failures	F9 v1.1 B1012 (Jan), F9 v1.1 B1015 (Apr) — both at CCAFS LC-40
Top landing outcomes, 2010-06-04 to 2017-03-20	No attempt (10), Success drone ship (6), Failure drone ship (5)

All Launch Site Markers



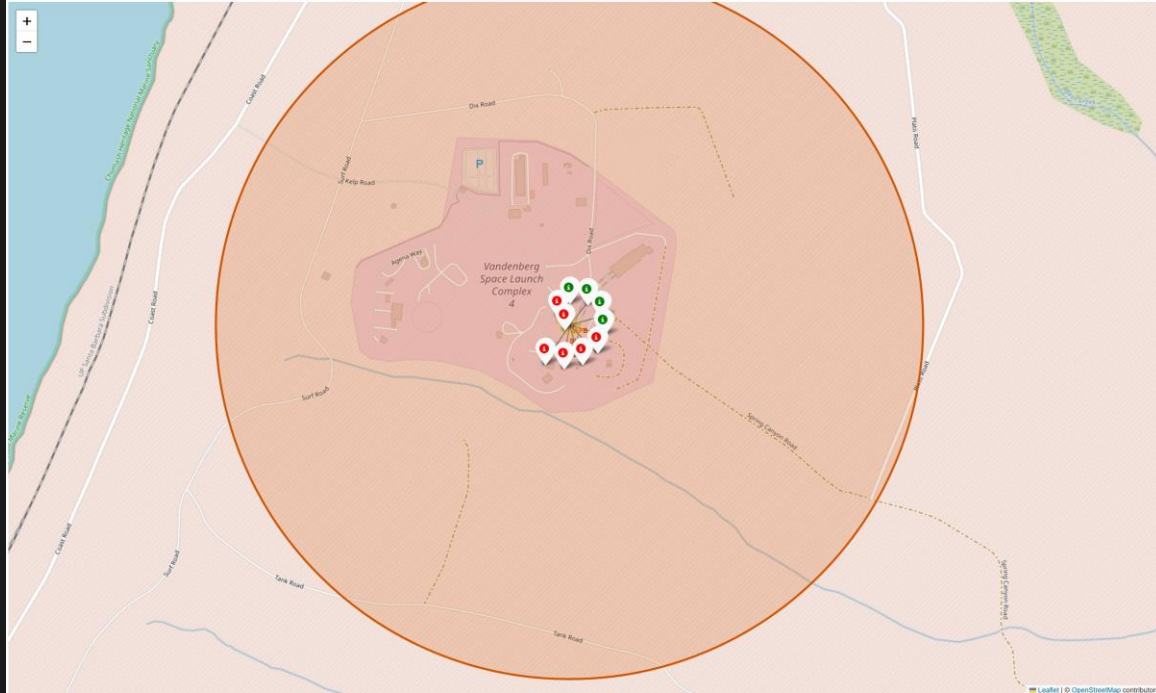
- All 4 launch sites are plotted with markers and labels
- All sites sit close to the Equator, which gives a rotational speed boost to orbit
- All sites sit close to the coast, reducing risk to populated areas from launch failures

Color-Coded Launch Outcomes (KSC LC-39A (13))



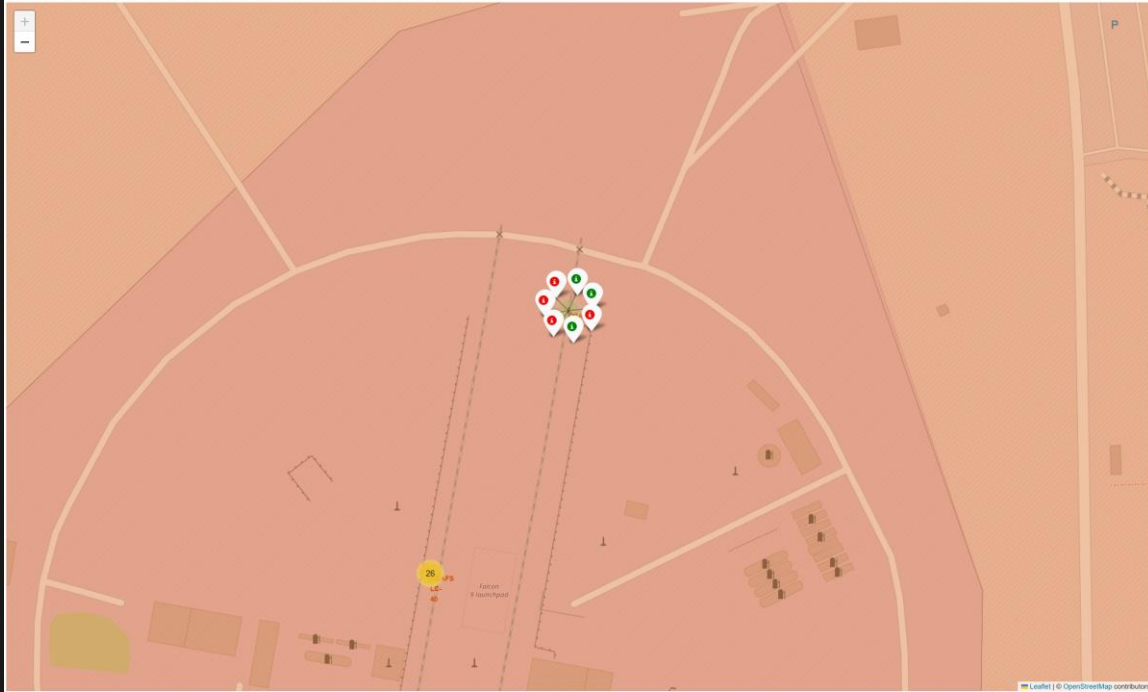
- Green markers = successful landing, red = failed landing
- Marker clustering makes per-site success patterns easy to scan visually
- KSC LC-39A clearly shows a majority-green cluster, consistent with its high success rate

Color-Coded Launch Outcomes (VAFB SLC4E (10))



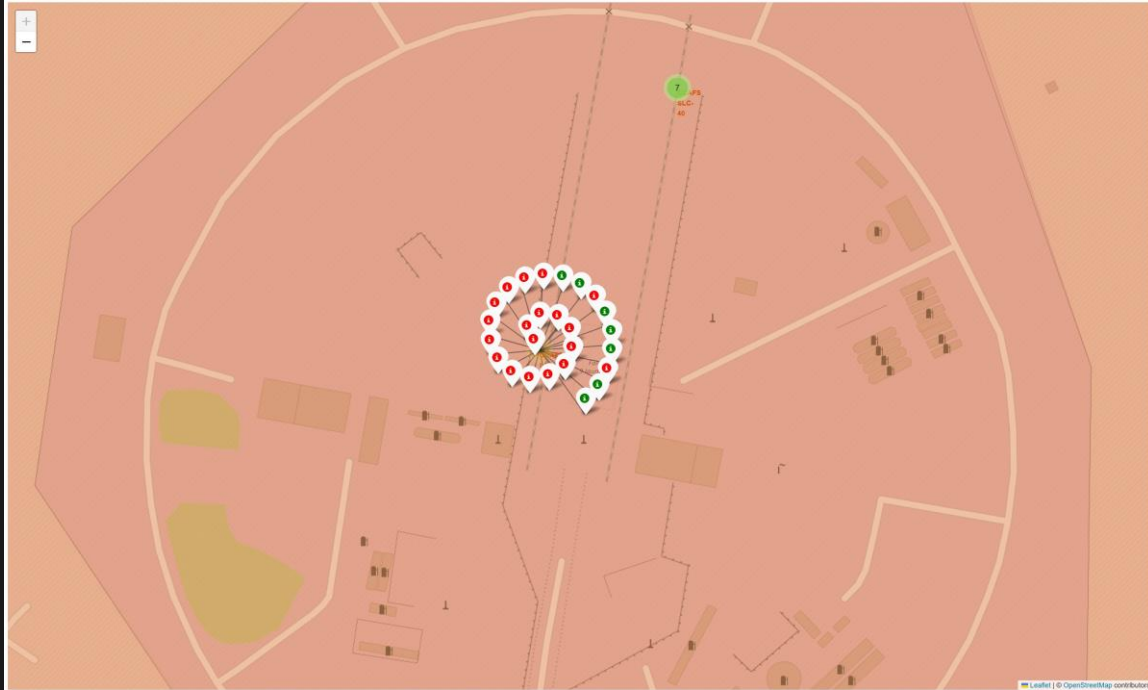
- Green markers = successful landing, red = failed landing
- Marker clustering makes per-site success patterns easy to scan visually
- KSC LC-39A clearly shows a majority-green cluster, consistent with its high success rate

Color-Coded Launch Outcomes (CCAFS SLC40 (7))



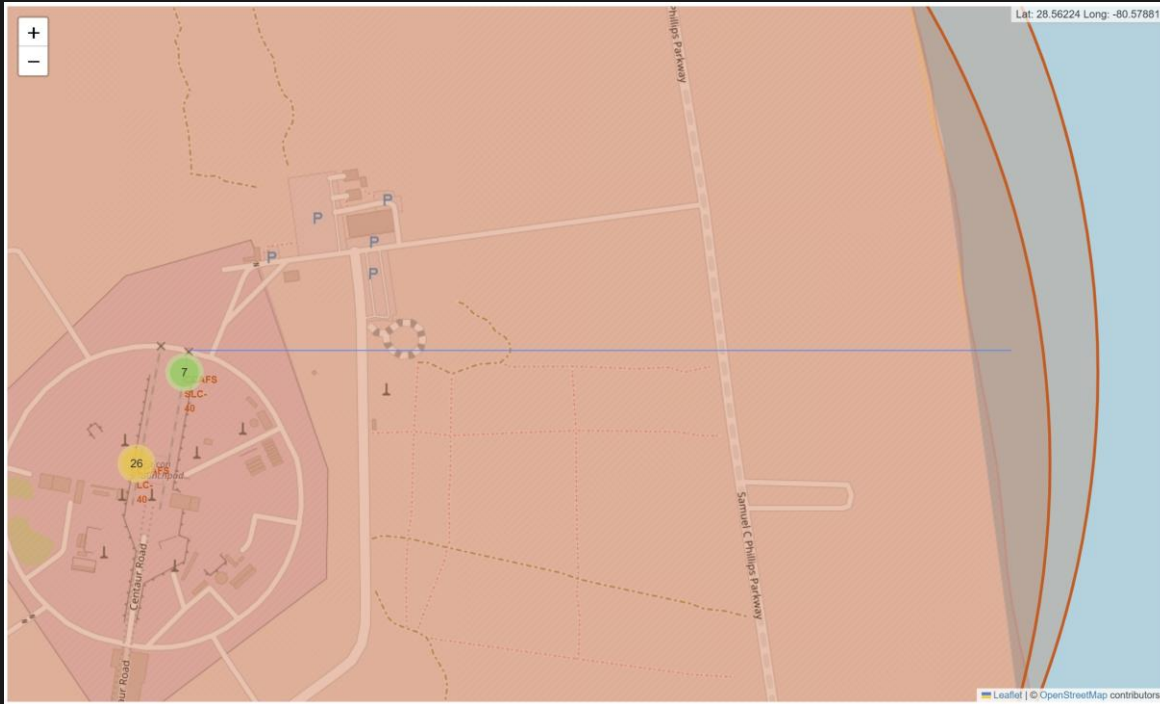
- Green markers = successful landing, red = failed landing
- Marker clustering makes per-site success patterns easy to scan visually
- KSC LC-39A clearly shows a majority-green cluster, consistent with its high success rate

Color-Coded Launch Outcomes (CCAFS SLC40 (26))



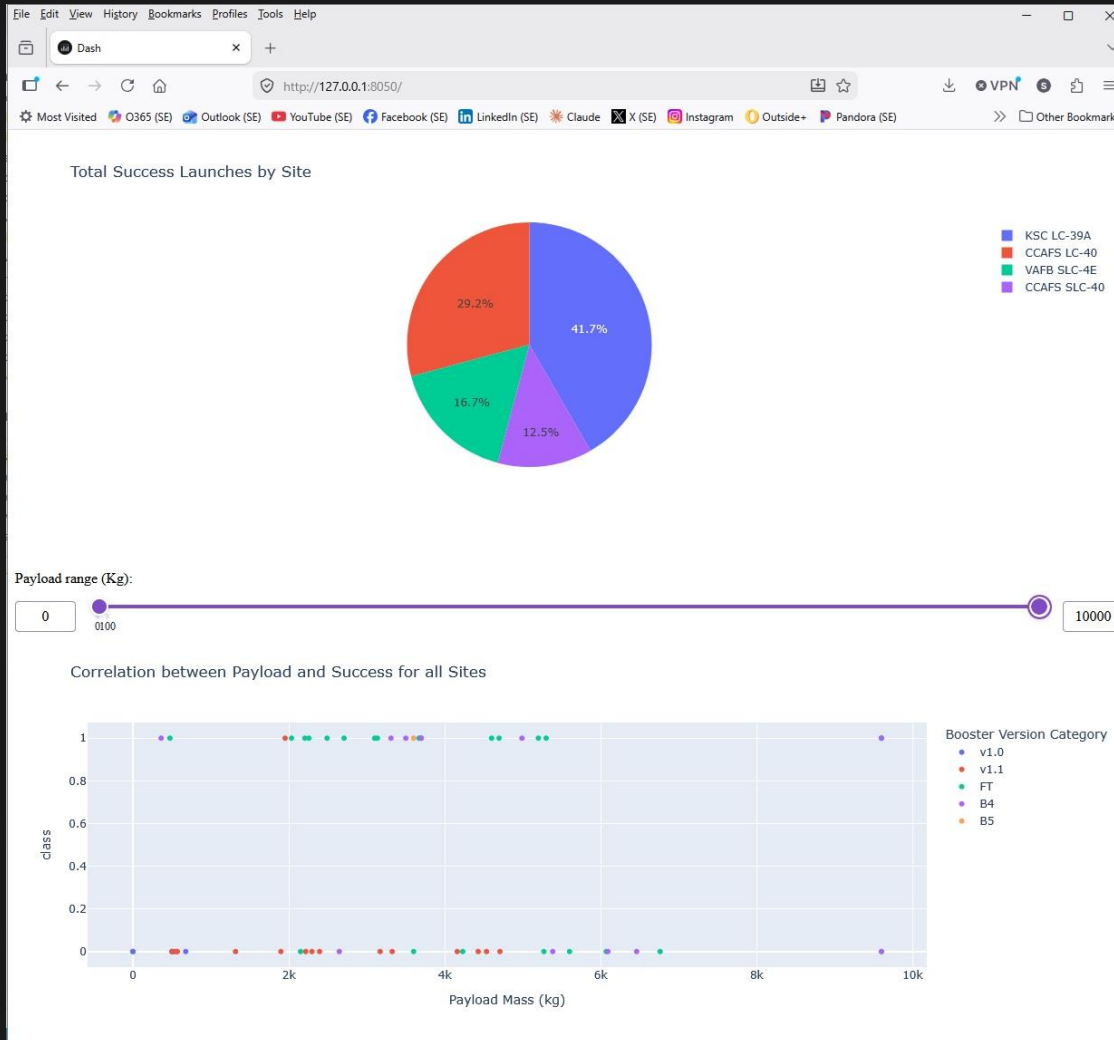
- Green markers = successful landing, red = failed landing
- Marker clustering makes per-site success patterns easy to scan visually
- KSC LC-39A clearly shows a majority-green cluster, consistent with its high success rate

Proximity Analysis



- Distance lines drawn from the launch site to nearby railway, highway, coastline, and city
- Confirms launch sites are sited away from dense population centers, favoring closeness to the coast
- A failed rocket at high speed could travel the railway/highway distance within seconds
- Supports the safety rationale behind launch site placement

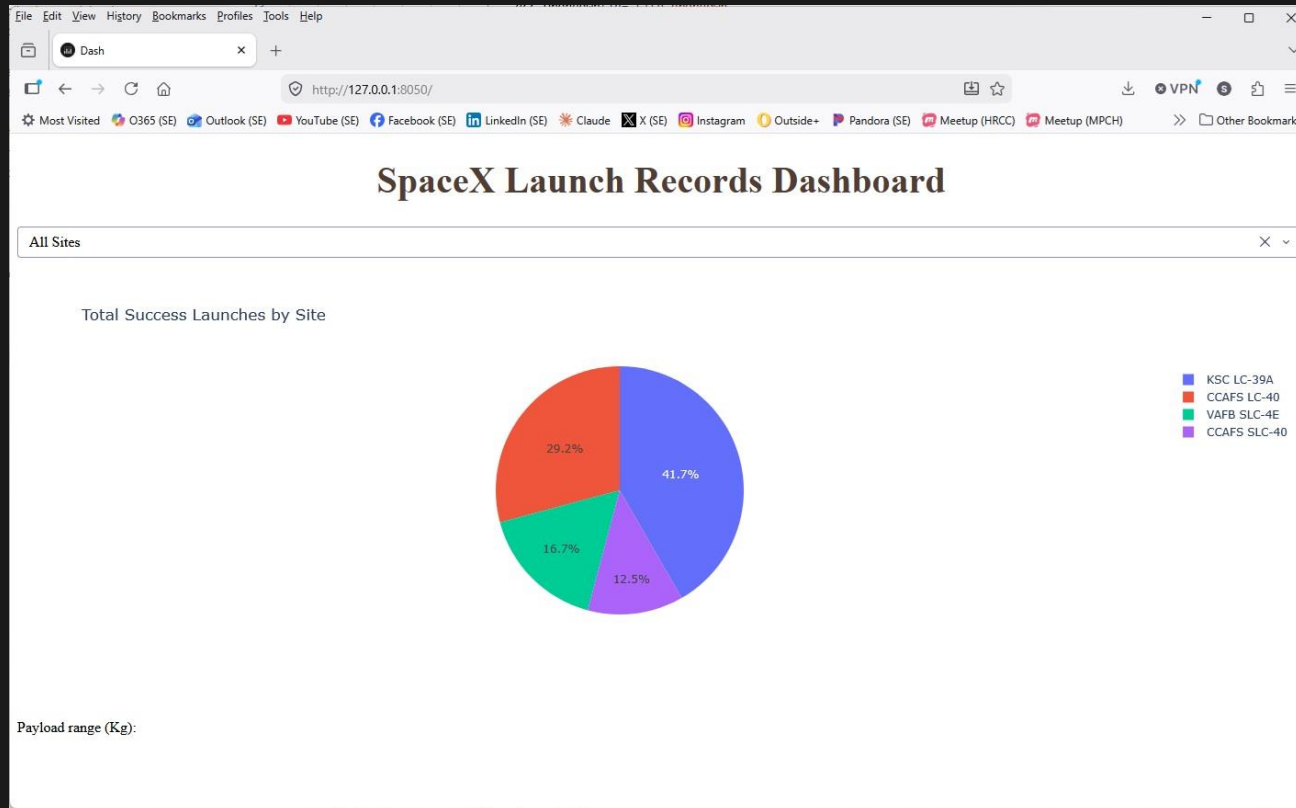
Dashboard



An interactive dashboard was developed to provide further visual analysis.

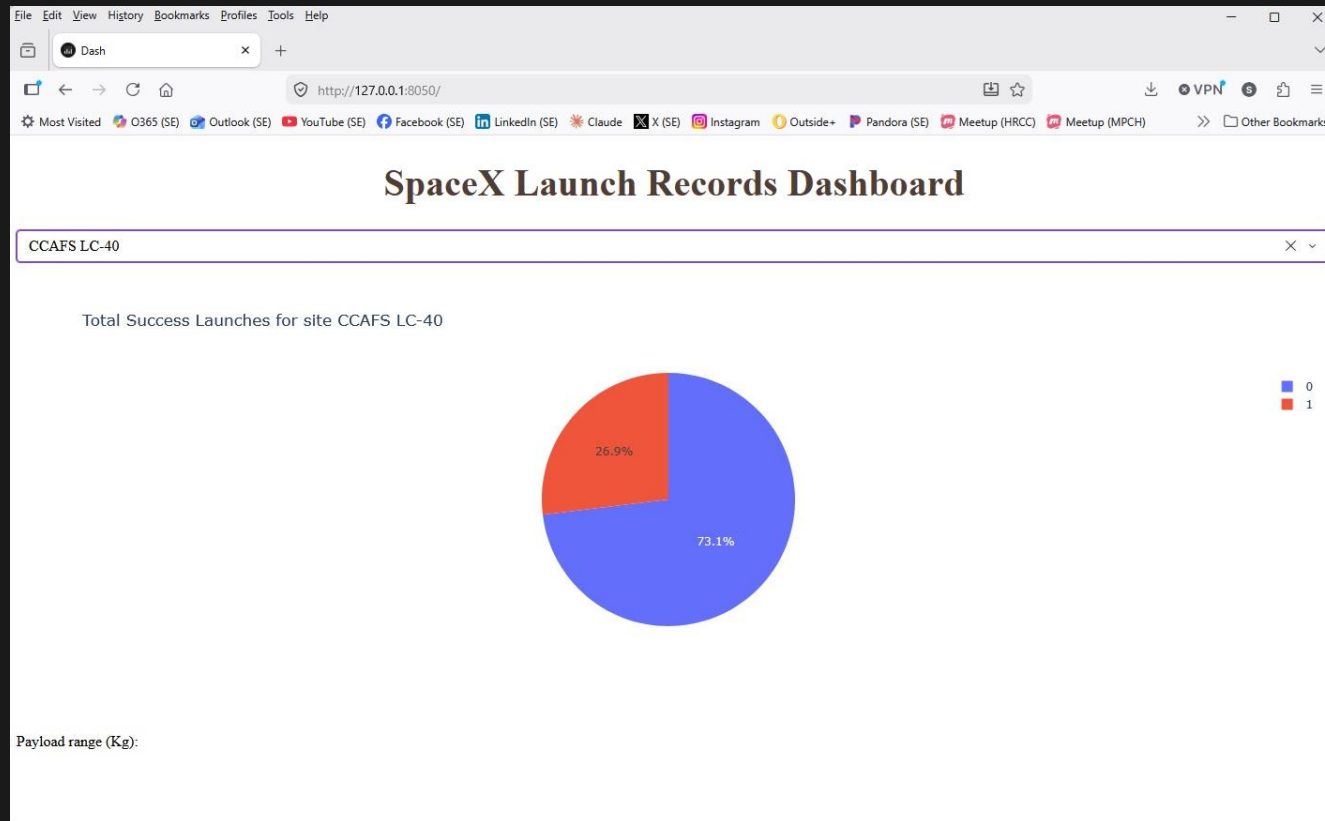
- Dropdown lets the user pick a single launch site or All Sites
- Payload range can be adjusted by means of a slider control

Dash App — Success Pie Chart (All Sites)



- KSC LC-39A accounts for the largest share (41.2%) of successful launches
- CCAFS SLC-40 follows at 23%, then VAFB SLC-4E at 21.4%, then CCAFS LC-40 at 14.4%

Dash App — Success Pie Chart (Single Site)



- Selecting a single site shows its own success vs. failure split
- KSC LC-39A: 10 successes vs. 3 failures — a 76.9% success rate

Dash App — Payload vs. Outcome Scatter

- Payload range slider filters the scatter plot live
- Points colored by booster version category (v1.0, v1.1, FT, B4, B5)
- Payloads roughly between 2,000 and 5,500 kg show the strongest success rate

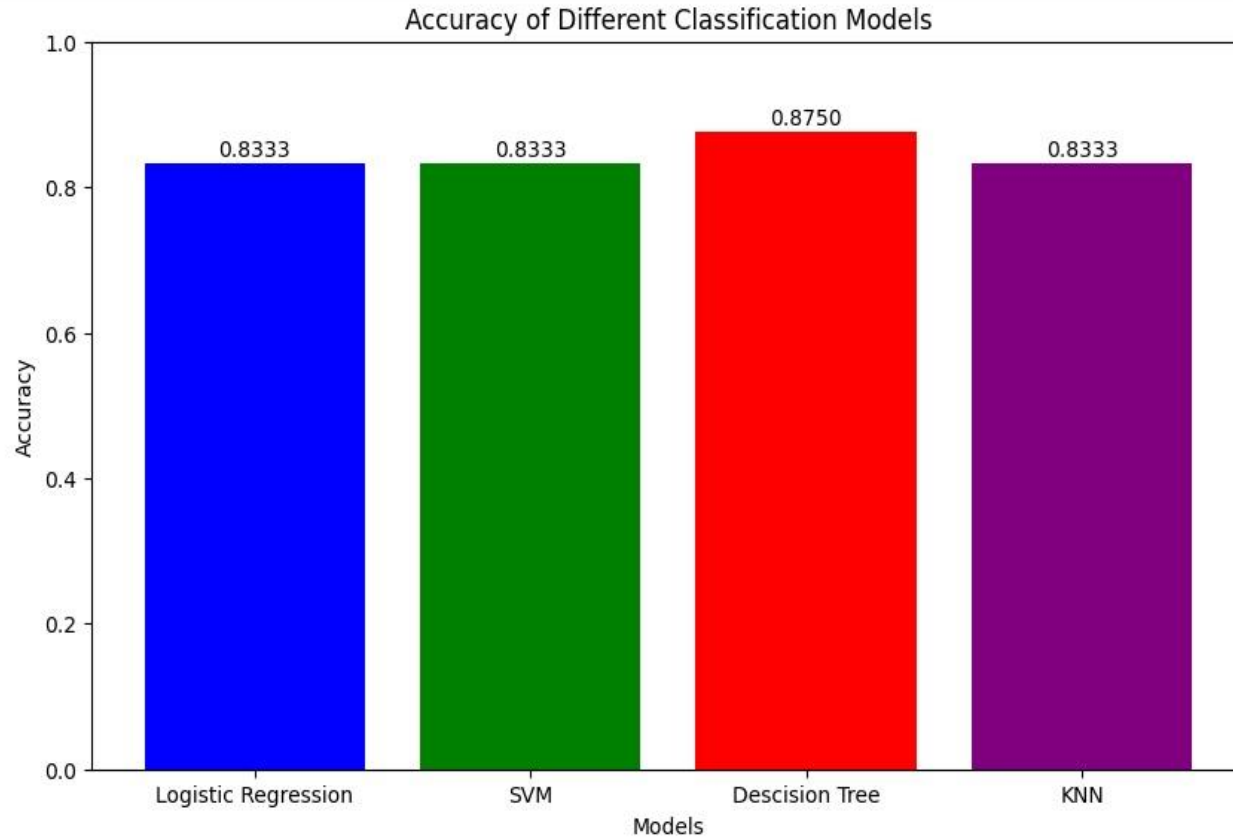


Predictive Analysis — Model Comparison

Model	Best CV Accuracy (GridSearchCV)	Test Set Accuracy
Logistic Regression	0.846429	0.8333
SVM	0.848214	0.8333
Decision Tree	0.889286	0.8888
KNN	0.848214	0.8333

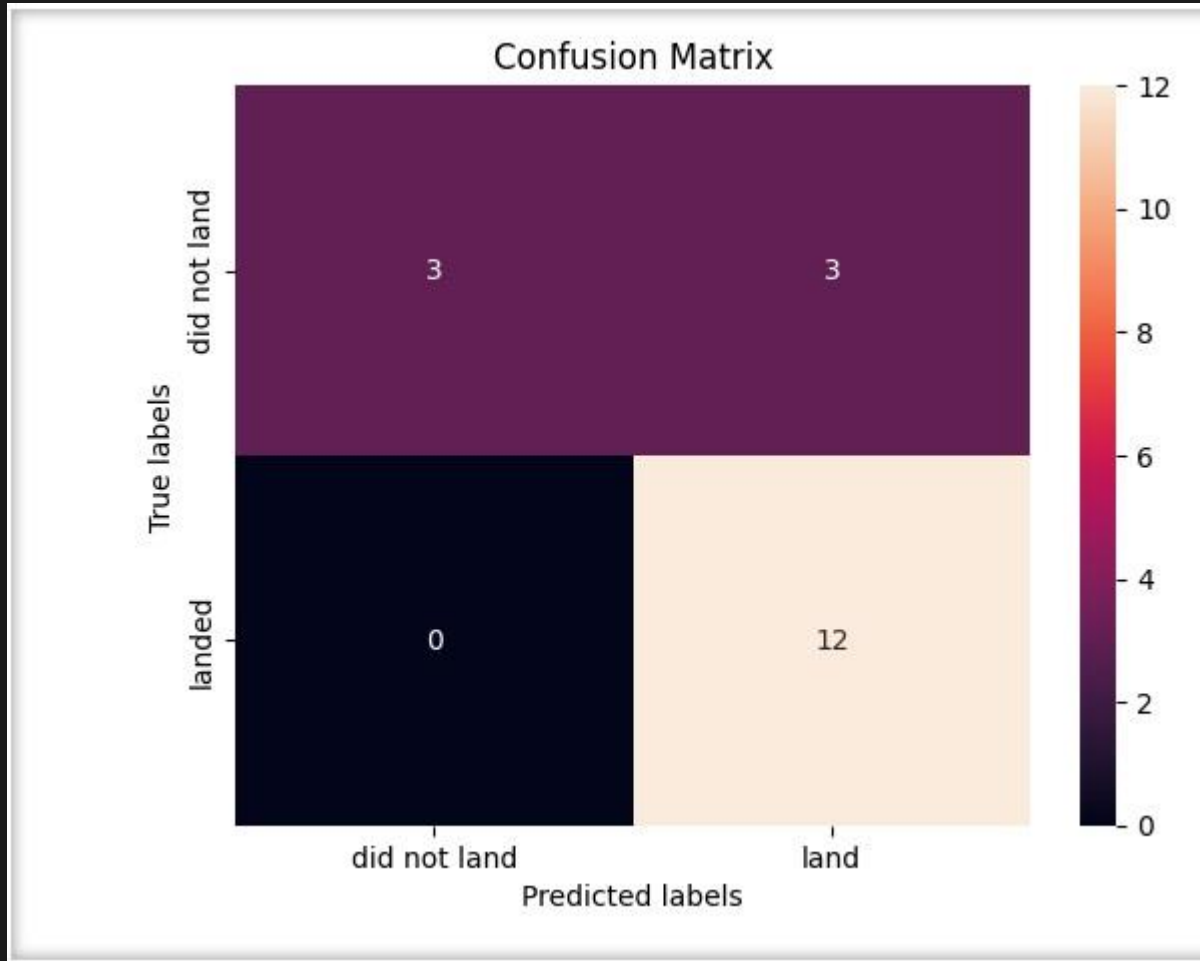
All four models were tuned with GridSearchCV (cv=10). On cross-validated accuracy, the Decision Tree classifier scored highest at 88.9%, ahead of SVM and KNN (84.8%) and Logistic Regression (84.6%). Test-set scores were similar across models due to the small held-out sample (18 records), so the cross-validated score is the more reliable comparison.

Best Performing Model



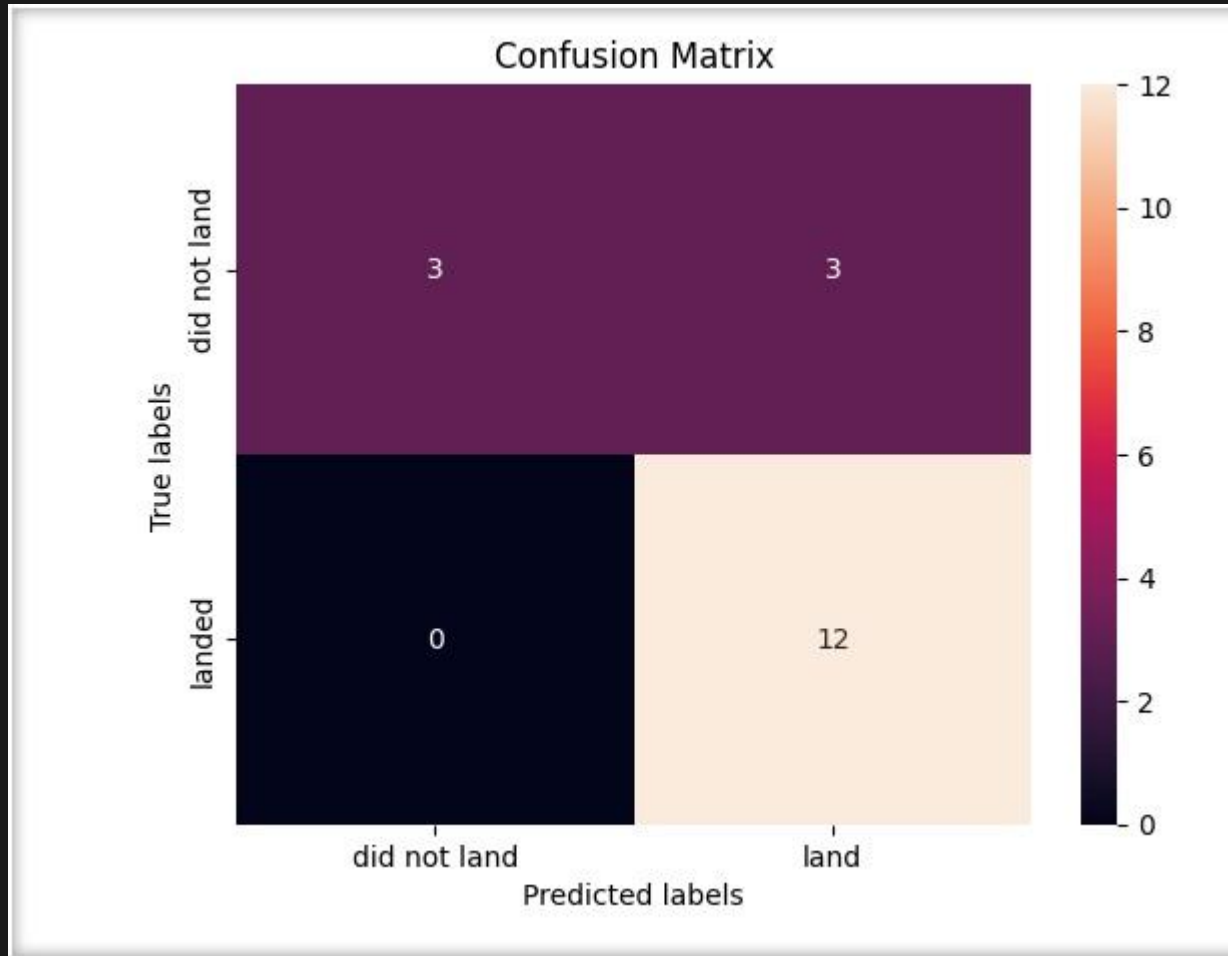
- Decision Tree achieves the highest accuracy (0.8750) of the four tuned models
- Logistic Regression, SVM, and KNN tie at 0.8333 on this comparison
- Decision Tree is selected as the best model for predicting first-stage landing

Confusion Matrix — Logistic Regression



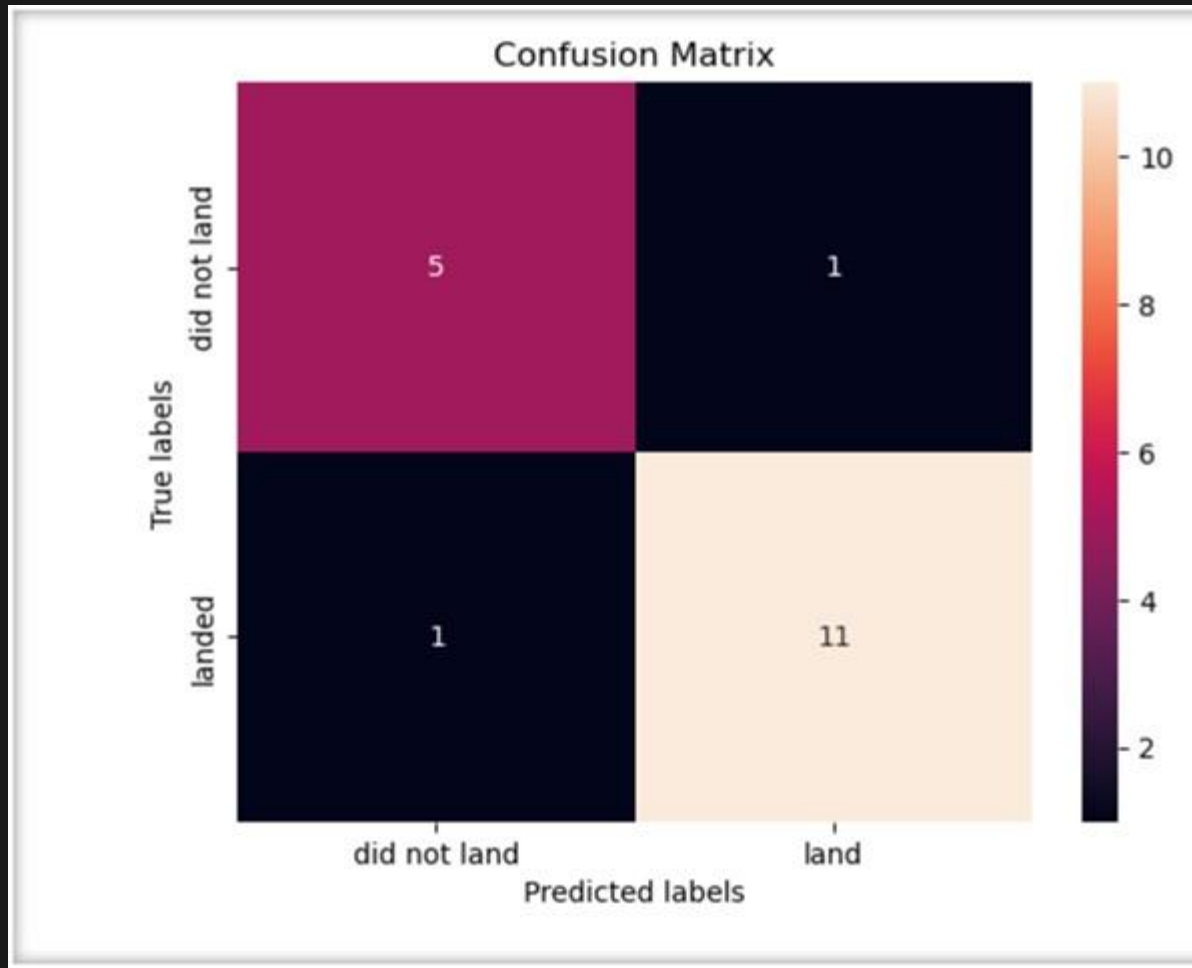
- Model correctly identifies all 12 successful landings (no false negatives)
- 3 of 6 failed landings are misclassified as successes (false positives)
- False positives are the dominant error mode for this model

Confusion Matrix — SVM



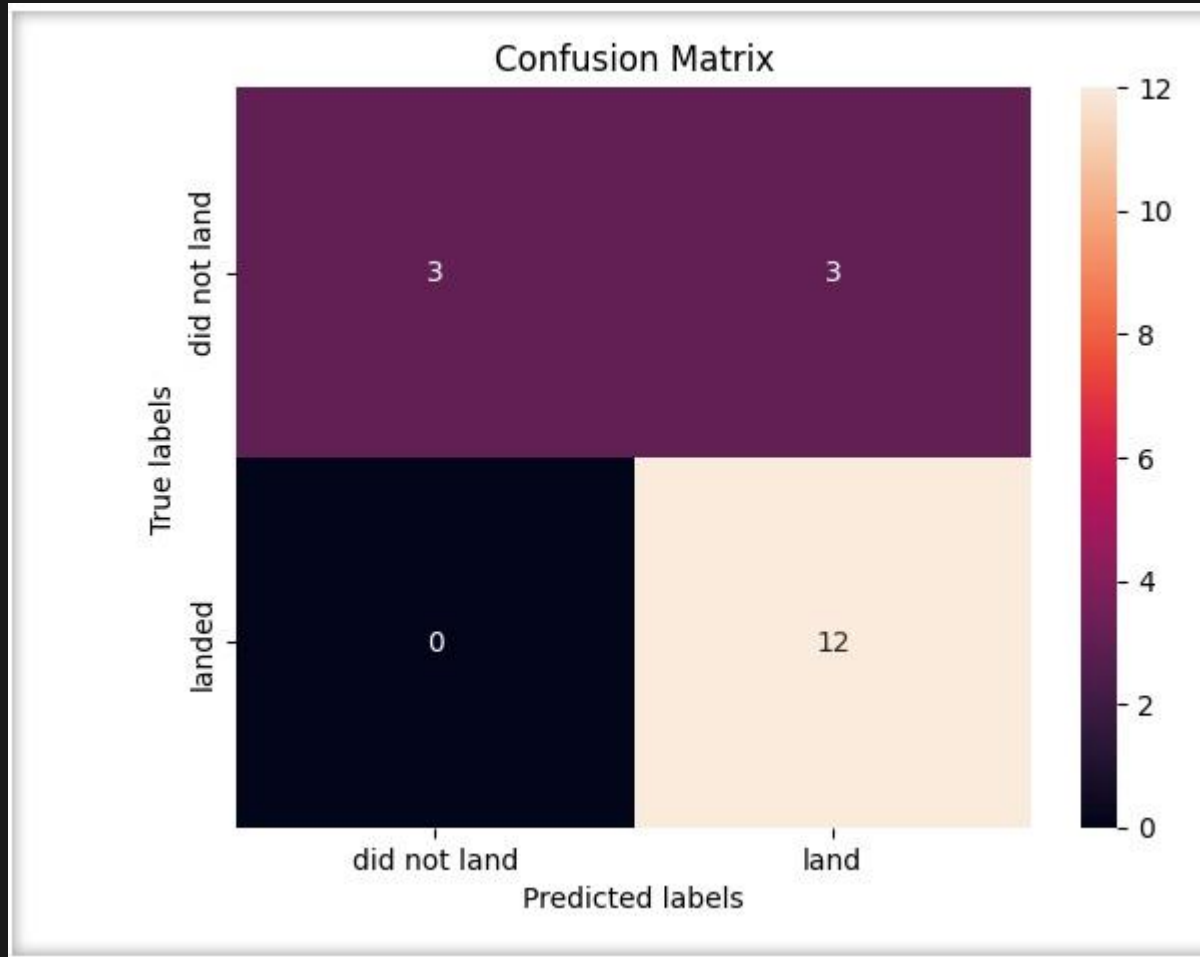
- Model correctly identifies all 12 successful landings (no false negatives)
- 3 of 6 failed landings are misclassified as successes (false positives)
- False positives are the dominant error mode for this model

Confusion Matrix — Decision Tree



- Model correctly identifies 11 successful landings
- Model correctly identifies 5 failed landings
- 1 of 6 failed landings are misclassified as successes (false positives)
- 1 of 6 successful landings are misclassified as failures (false negatives)

Confusion Matrix — KNN



- Model correctly identifies all 12 successful landings (no false negatives)
- 3 of 6 failed landings are misclassified as successes (false positives)
- False positives are the dominant error mode for this model

Discussion

- Landing success is driven by a combination of factors, not any single variable: launch site, payload mass, and orbit type all interact
- The steady year-over-year improvement in success rate reflects SpaceX's iterative engineering process rather than a one-time fix
- Mid-range payloads (roughly 2,000-5,500 kg) and orbits like GEO/HEO/SSO are the most reliable bets for a successful landing
- All four classifiers perform similarly on raw test accuracy because of the small dataset (90 launches); cross-validation is essential to distinguish them
- False positives (predicting a landing when none occurred) are the main error mode — a meaningful limitation if this model were used to estimate launch cost in advance

Conclusion

- The Decision Tree model is the best-performing algorithm for predicting Falcon 9 first-stage landing outcomes (87.5% cross-validated accuracy)
- Launch site, payload mass, and orbit type all materially affect landing success and should be treated as core predictive features
- Landing success has trended upward every year since 2013, reflecting SpaceX's maturing recovery process
- KSC LC-39A is the strongest-performing launch site, with a 76.9% success rate
- Orbits ES-L1, GEO, HEO, and SSO have a 100% observed success rate in this dataset
- Because the first stage is reusable when it lands successfully, this kind of model gives a data-driven way to estimate the cost of a future SpaceX launch

Index / Appendix

Project repository:

GitHub: <https://github.com/StephenElmer/ibm-applied-data-science-capstone>

Stage notebooks and apps:

Data Collection — API: [01-1-jupyter-labs-spacex-data-collection-api.ipynb](#)

Data Collection — Web Scraping: [01-2-jupyter-labs-webscraping.ipynb](#)

Data Wrangling: [01-3-labs-jupyter-spacex-Data wrangling.ipynb](#)

EDA with SQL: [02-1-jupyter-labs-eda-sql-coursera_sqlite.ipynb](#)

EDA with Visualization: [02-2-edadataviz.ipynb](#)

Interactive Visual Analytics — Folium: [03-1-lab_jupyter_launch_site_location.ipynb](#)

Interactive Dashboard — Plotly Dash: [03-2-spacex_dash_app.py](#)

Machine Learning Prediction: [04-1-SpaceX_Machine_Learning_Prediction_Part_5.ipynb](#)

With thanks to IBM and Coursera for the Applied Data Science Capstone curriculum.